

HANG SENG INDEX

Social Media Prediction Vote Analysis & Trading Strategy Report

Period: 2022-02-23 to 2025-03-12

747

552

47.8%

+6.5%

Trading Days

Vote Days
(Coverage 73.9%)

Crowd Accuracy

B&H, Total Return
(Sharpe 0.21)

Executive Summary

Social media pre-market votes on HSI are a powerful **contrarian indicator**: the majority is directionally correct only **47.8%** of the time. Pure sentiment strategies (S3 Contrarian, S4 Extreme Contrarian) already beat Buy & Hold (+6.5%, Sharpe 0.21). Adding SMA technical filters then transforms these strategies dramatically, **S4+SMA delivers +31.6% (Sharpe 5.64, Max Drawdown just -3.4%)**. The final recommendation integrates both layers: sentiment identifies when the crowd is wrong; the SMA filter confirms which direction the market is actually trending

Report Structure

This report is organised to analyze data and build the trading strategy step by step - from raw data through base strategies through technical enhancement, concluding with a single combined recommendation that uses both sentiment and technical signals together.

#	Section	Description
1	Data Overview & Methodology	Dataset statistics, backtest rules, coverage analysis
2	Key Analytical Insights	Data-driven findings about crowd behaviour and HSI returns
3	Strategy Definitions	Base sentiment strategies explained
4	Base Strategy Results	Backtest performance - sentiment signals only
5	Technical Indicator Filters	SMA, RSI, MACD as confirmation gates on top of sentiment
6	Filtered Strategy Result	How technical filters transform S3 & S4 performance
7	Final Recommendation	Combined sentiment + technical strategy - the complete system

1. Data Overview & Methodology

The dataset covers HSI daily OHLC prices from 2022-02-23 to 2025-03-12 (747 trading days). Pre-market Up/Down vote fractions from social media are available for 552 days (73.9%). The crowd is systematically bullish-biased: average Up vote = 54.1%, Down = 46.0%. Bullish-majority days: 350 (63%); Bearish-majority days: 202 (37%).

Metric	Value	Metric	Value
Total Trading Days	747	Vote Coverage	73.9%
Avg Up Vote	54.1%	Avg Down Vote	46.0%
Directional Accuracy	47.8%	B&H Total Return	+6.5%
B&H CAGR	+2.1%	B&H Sharpe Ratio	0.21

Backtest Rules:

- Using signal from pre-market votes before Day T open to trade day T
- Entry at HSI Open, Exit at HSI Close (same day intraday)
- Commission: 0.05% per trade
- Initial capital: HK\$100,000. No leverage. No overnight positions

HSI Price and Social Media Sentiment Overview

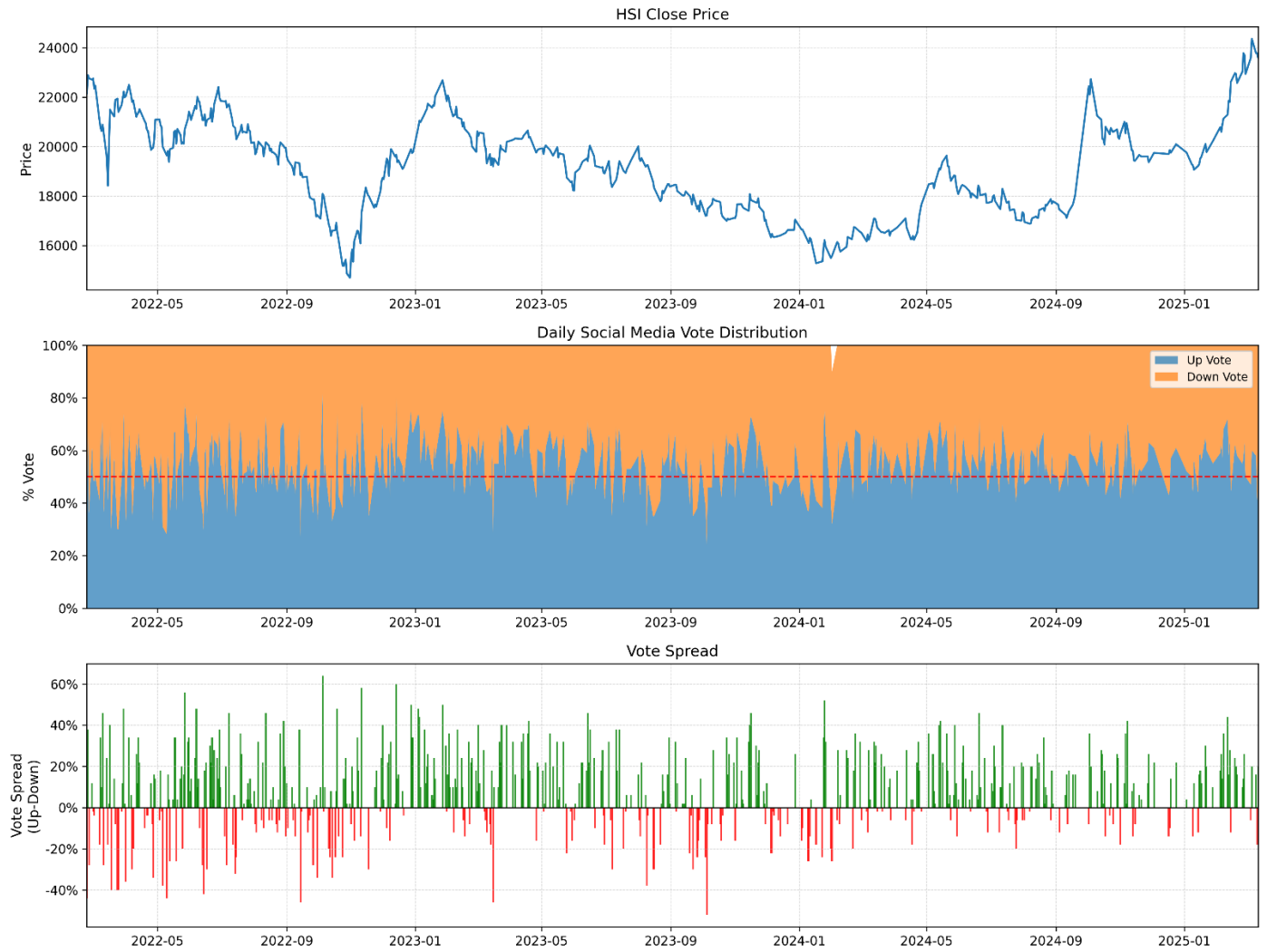


Figure 1: Overview: HSI Close Price, Daily Social Media Vote Distribution, Daily Vote Spread

Prediction Vote Signal Accuracy and Statistical Analysis

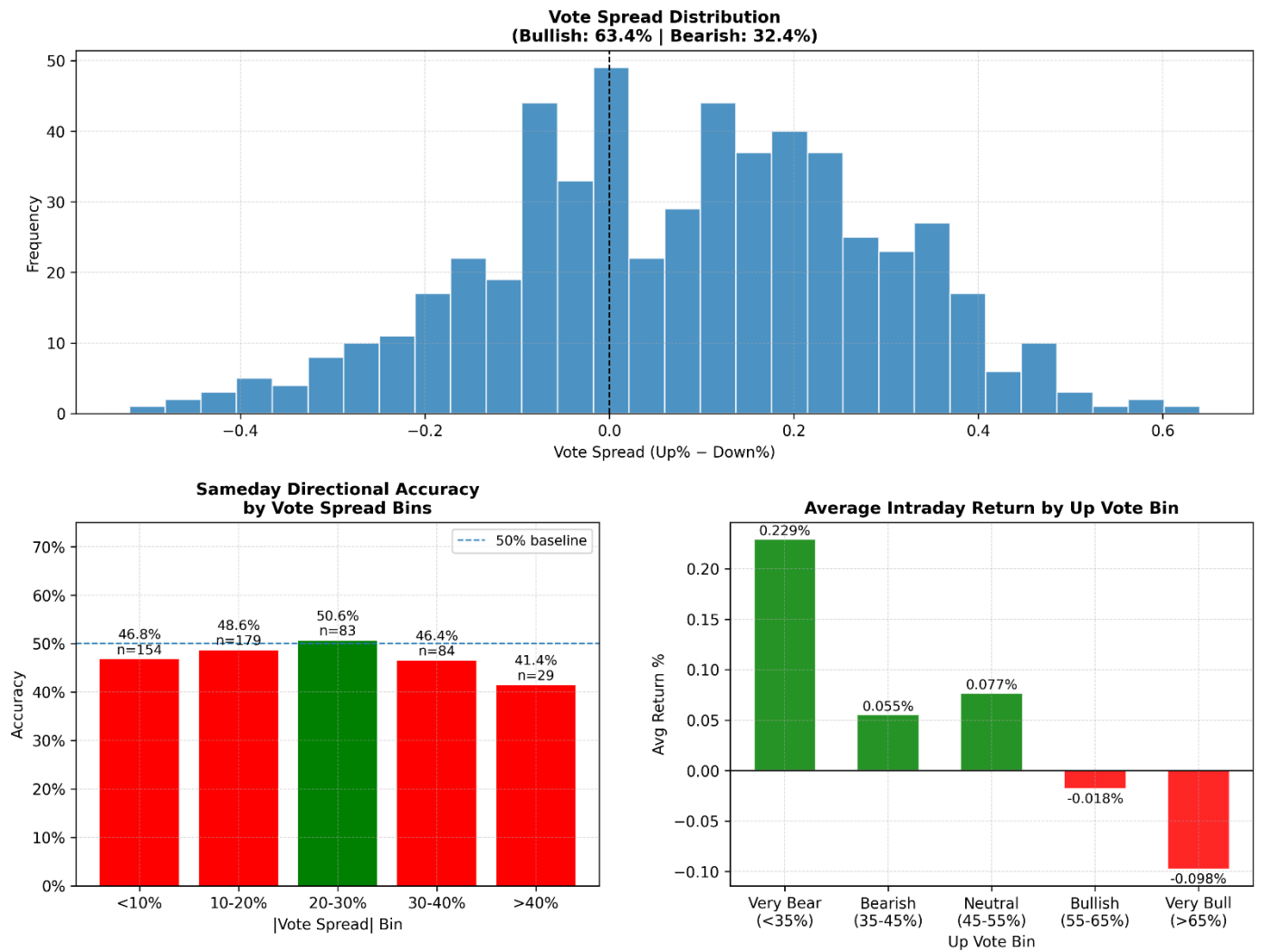


Figure 2: Accuracy Analysis: Vote Spread Distribution, Direction Accuracy by Conviction level, Avg HSI Return by Up vote bucket

2. Key Analytical Insights

■ Crowd is a Consistent Contrarian Indicator

Overall accuracy is 47.8%, below the 50% baseline across the full period. Social media voters consistently misread short-term HSI direction, consistent with the "dumb money" sentiment effect documented in equity markets globally.

■ Higher Conviction = More Reliably Wrong

The largest conviction bucket (>40% spread) has accuracy of only 41.4% (n=29 days). When the crowd agrees most strongly, the market is most likely to disappoint them, this pattern makes extreme conviction the most actionable signal.

■ Very Bullish Crowd Predicts Negative Returns

Average intraday return declines monotonically as bullishness rises: Very Bear days (<35% Up) have average return +0.229%, while Very Bull day (>65% Up) have average return -0.098%. An over-bullish crowd has already exhausted near-term buying pressure.

■ Extreme Sentiment is the Most Actionable Signal

When Up votes exceed 60% (153 days): market moved DOWN 55.6% of the time (avg return -0.133%). When Down votes exceed 60% (43 days): market moved UP 41.9% of the time. Extreme bulls are more reliably wrong, consistent with overconfidence bias in retail bullish sentiment.

■ Technical Filters Unlock Dramatically Better Risk-Adjusted Returns

Adding SMA20 to S3 Contrarian lifts Sharpe from 0.31 to 3.76 (10x improvement) while cutting max drawdown from -22.0% to -10.5%. For S4 Extreme Contrarian: Sharpe 0.61 to 5.64, max drawdown -16.6% to -3.4%. The filter prevents contrarian trades from fighting strong momentum

3. Strategy Definitions

Four base strategies are evaluated. S3 and S4, both profitable and with positive Sharpe, are selected for technical filter enhancement in Section 5.

Strategy	Signal Rule	Side	Trades
S1.Follow Majority (Always)	Long if Up vote > Down vote Short if Down vote > Up vote	Long/Short	552
S2.Follow Majority (High Conviction)	Long if (Up – Down) > 0.15 Short if (Down – Up) > 0.15 Flat otherwise	Long/Short	278
S3.Contrarian (Always)	Long if Up vote < Down vote Short if Down vote < Up vote	Long/Short	552
S4.Extreme Contrarian	Long if Down vote > 0.6 Short if Up vote > 0.6 Flat otherwise	Long/Short	196

- S3 & S4 progress to Section 5 (technical filter).

4. Base Strategy Result (Sentiment Signal Only)

Performance of all four strategies over 552 voting days with HK\$100,000 initial capital and 0.05% commission per trade. These results use sentiment signals only, no technical filters. Note that even the best base strategies here will be significantly improved in Section 5 once filters are applied.

Strategy	Trade	Win Rate	Total Ret	CAGR	Sharpe	Max DD	Avg Ret
S1.Follow Majority (Always)	552	47.1%	-52.3%	-21.6%	-1.53	-53.8%	-0.1255%
S2.Follow Majority (High Conviction)	278	46.8%	-26.5%	-9.6%	-1.26	-26.6%	-0.1025%
S3.Contrarian (Always)	552	50.7%	+9.9%	+3.1%	0.31	-22.0%	+0.0255%
S4.Extreme Contrarian	196	51.5%	+8.1%	+2.6%	0.61	-16.6%	+0.0477%

- S3 (Blue) = Highest Return base strategy.
- S4 (Green) = Best Sharpe and lowest drawdown base strategy.
- Buy and Hold: Total Return: 6.5% | CAGR 2.1% | Sharpe 0.21

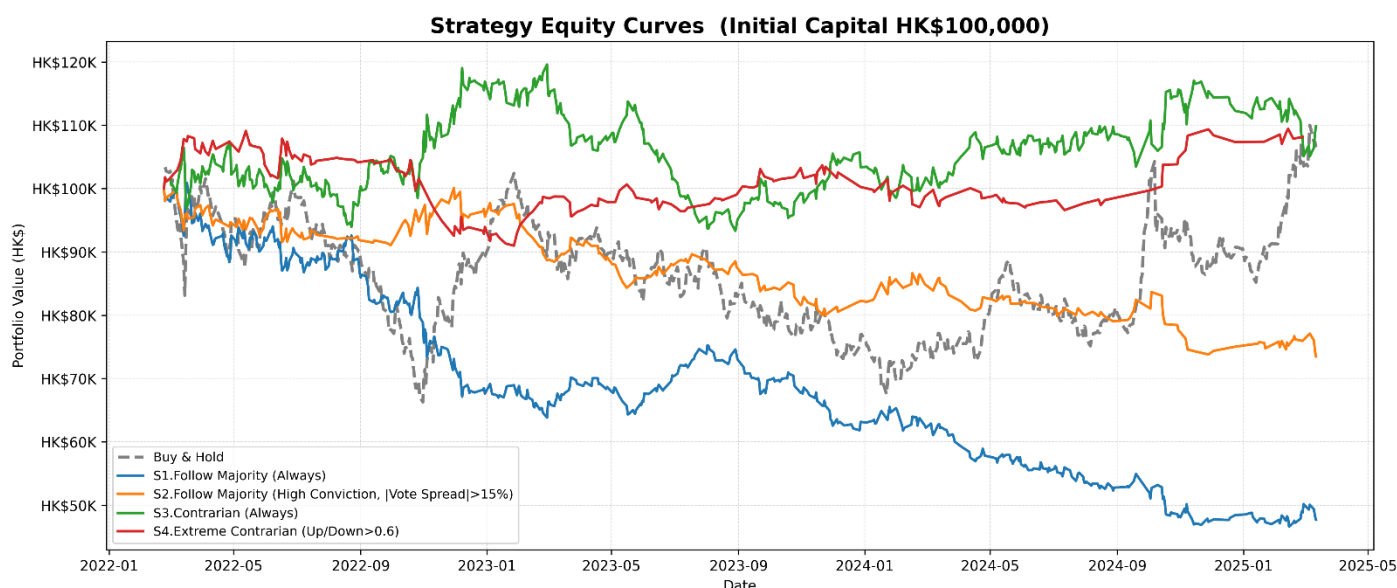


Figure 3: Equity curves - all 4 base strategies vs Buy & Hold. S1 and S2 destroy capital; S3 and S4 consistently beat B&H

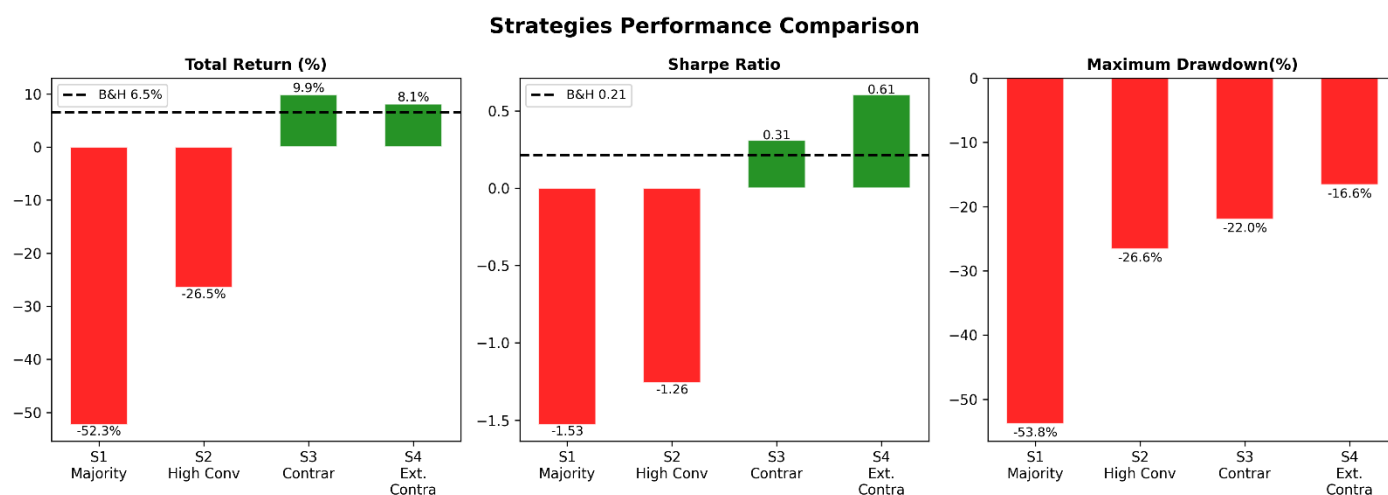


Figure 4: Total Return, Sharpe Ratio, and Max Drawdown across base strategies

Base Strategy Interpretation

S1.Follow Majority (Always) (Total Return -52.3%, Sharpe -1.53): 552 trades, win rate 47.1%. Always following the crowd destroys capital with -53.8% max drawdown.

S2 High Conviction (Total return -26.5%, Sharpe -1.26): Selective following crowd on high-conviction days (278 trades) reduces losses compared to S1 but doesn't help avoiding negative return and Sharpe, with -26.6% max drawdown.

S3 Contrarian (+9.9%, Sharpe 0.31): First clearly profitable strategy. Fading the crowd on all 552 vote days. Win rate 50.7%, max drawdown -22.0%. Selected for filter enhancement.

S4 Extreme Contrarian (+8.1%, Sharpe 0.61): Best Sharpe in base group at 0.61. Only 196 trades on the most extreme conviction days. Win rate 51.5%, max DD -16.6%. Selected for filter enhancement.

5. Technical Indicator Filters

S3 and S4 are profitable on sentiment alone, but they still take trades against the prevailing price trend, which increases drawdown and noise. Technical indicators are added as a second decision layer: the sentiment signal identifies when to be contrarian; the indicator filter confirms whether the market regime supports that direction.

Indicator	Parameters	Bullish Signal	Bearish Signal	Filter Mode
SMA 20	20-day rolling mean on Close	Close > SMA20 => uptrend	Close < SMA20 => downtrend'	Agree
RSI 14	14-period EWM smoothing	RSI < 40 => oversold	RSI > 60 => overbought	Non-oppose
MACD	EMA12 - EMA26; Signal = EMA9	MACD > Signal => bull momentum	MACD < Signal => bear momentum	Agree
All 3 combined	Score = SMA + RSI + MACD; Range: -3 to +3	Score > 0 => majority bullish	Score < 0 => majority bearish	Agree

Filter Mode Rationale

Agree (SMA & MACD): These indicators always produce a clear ± 1 with no neutral zone. Trade only when the indicator fully agrees with the sentiment direction. Strictly reduces trade count, ensuring every trade has both sentiment AND technical confirmation.

Non-oppose (RSI): RSI has a neutral zone (40-60) where it has no directional opinion. "Agree" mode would wrongly block neutral-RSI days. "Non-oppose" only blocks when RSI actively contradicts e.g. long signal with overbought RSI (>60), allowing trades when RSI is neutral.

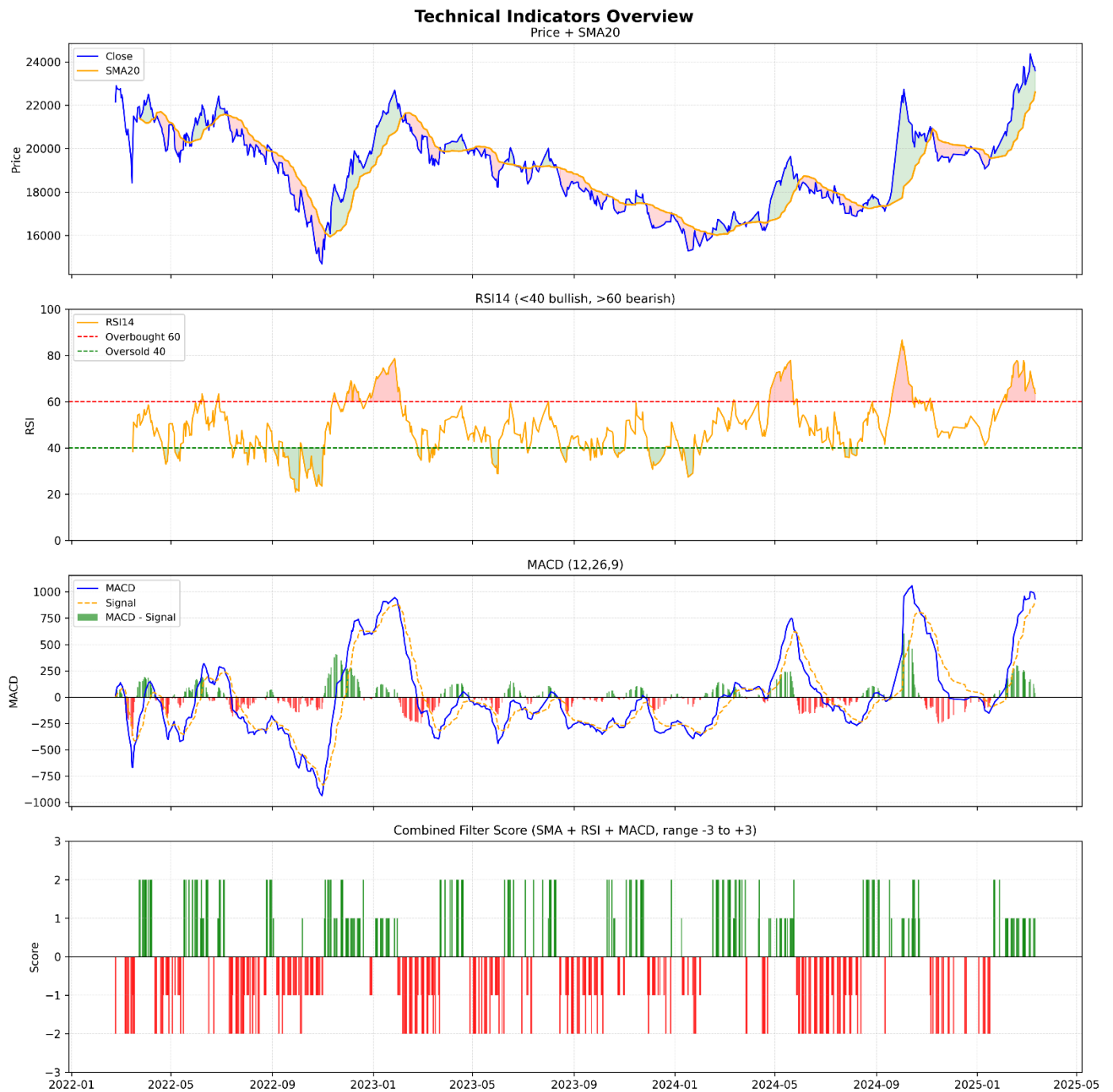


Figure 5: Technical indicators: Price+SMA20 with regime shading, RSI(14) with 40/60 zones, MACD(12,26,9) histogram, combined filter score (-3 to +3)

6. Filtered Strategy Results

Eight filtered variants are tested - four for each of S3 and S4. The improvements over the base strategies are substantial across all three key metrics: Return, Sharpe, and Max Drawdown.

Strategy	Trades	Win Rate	Total Ret	CAGR	Sharpe	Max DD	Avg Ret
Contra	552	50.7%	+9.9%	+3.1%	0.31	-22.0%	+0.0255%
Ext Contra	196	51.5%	+8.1%	+2.6%	0.61	-16.6%	+0.0477%
Contra + SMA	229	59.8%	+109.4%	+27.5%	3.76	-10.5%	+0.3332%
Contra + RSI	491	48.5%	-29.0%	-10.6%	-0.82	-38.2%	-0.0624%
Contra + MACD	246	54.9%	+68.1%	+18.6%	2.47	-10.7%	+0.2214%
Contra + All Filters	208	57.7%	+82.8%	+22.2%	3.31	-10.9%	+0.3008%
Ext Contra + SMA	70	64.3%	+31.6%	+10.4%	5.64	-3.4%	+0.3994%
Ext Contra + RSI	187	49.7%	+2.3%	+0.7%	0.25	-18.1%	+0.0198%
Ext Contra + MACD	76	56.6%	+18.4%	+6.2%	3.21	-5.3%	+0.2287%
Ext Contra + All Filter	65	61.5%	+25.9%	+8.8%	5.13	-4.0%	+0.3612%

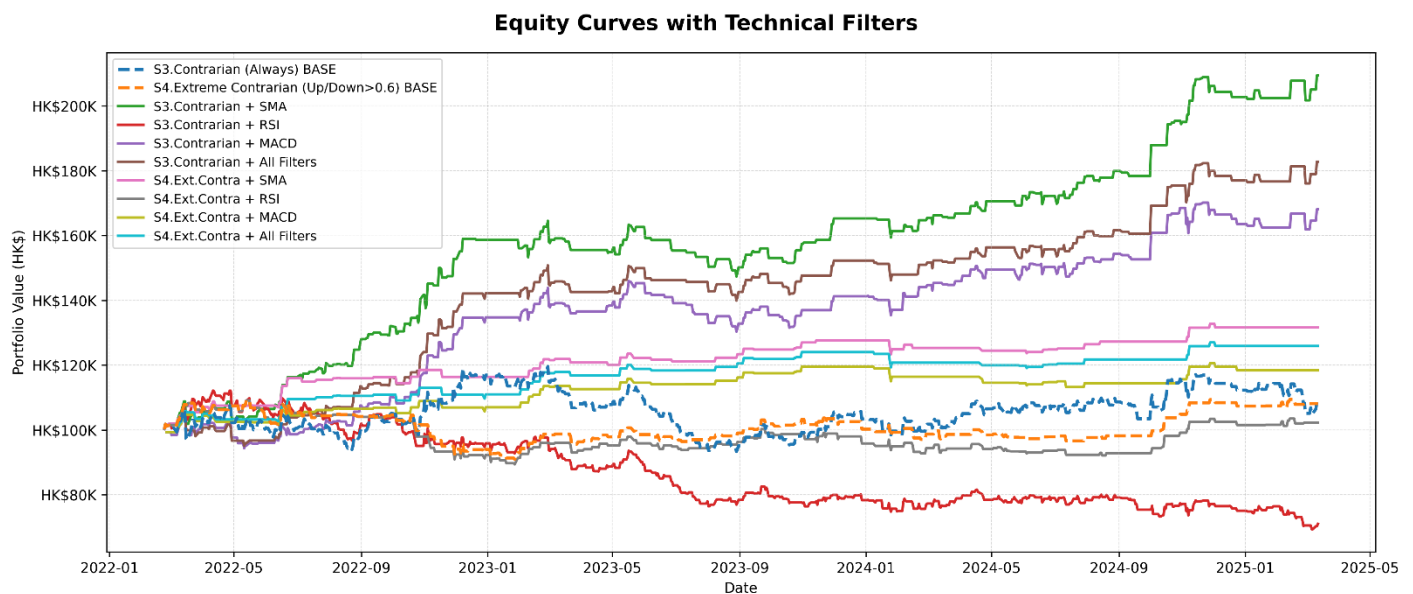


Fig 6: Equity curves - base vs filtered variants. Flat periods show cash-holding when filter blocks the trade

Performance Comparison: Base vs Technical Filtered Strategies

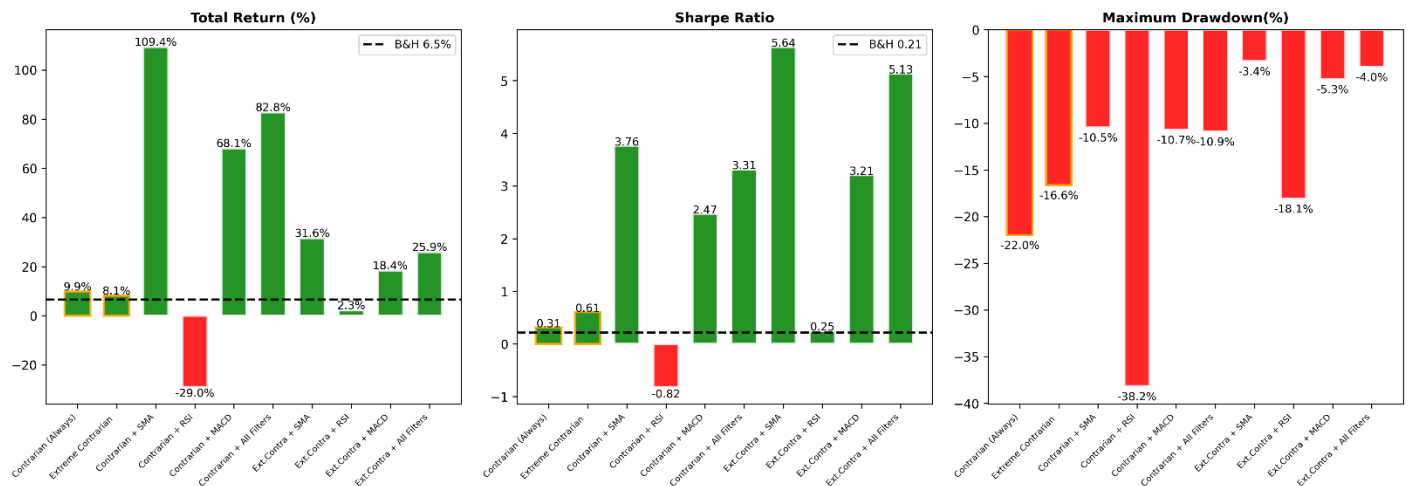


Fig 7: Full performance comparison all base and filtered variants. Orange-bordered = base strategies; white-bordered = filtered.

Filter by Filter Breakdown

SMA Filter - Best Regime Gate: S3+SMA: Total Return +109.4% | Sharpe 3.76 | Max Drawdown -10.5% (229 trades). S4+SMA: Total Return +31.6% | Sharpe 5.64 | Max DD -3.4% (70 trades). SMA enforces that Contrarian trades work best within the prevailing trend.

MACD Filter - Strong Momentum Confirmation: S3+MACD: Total Return +68.1% | Sharpe 2.47 (246 trades). S4+MACD: Total return +18.4% | Sharpe 3.21 (76 trades). MACD captures short-term momentum shifts, avoiding trades during strong continuation.

RSI Filter - Limited Standalone Value: S3+RSI: Total Return -29.0% | Sharpe -0.82 (491 trades). The non-oppose mode retains most trades, so RSI has limited selectivity alone. Therefore, it adds more value as part of the combined all filter than independently.

All Filters Combined - Maximum Selectivity: S3+All: Total Return +82.8% | Sharpe 3.31 | Max Drawdown -10.9% (208 trades). S4+All: Total Return +25.9% | Sharpe 5.13 | Max Drawdown -4.0% (65 trades). Rarest trades, but highest confidence, all three indicators plus extreme sentiment agree simultaneously

7. Final Recommendations

Combined Sentiment + Technical Filter Strategy

After evaluating 4 base sentiment strategies and eight technically-filtered variants, the best strategies combine both signals: sentiment tells us when the crowd is wrong; the technical filter confirms which direction the market is trending. Together, they produce a system that is far superior to either signal alone.

Primary Recommendation – S4. Extreme Contrarian + SMA

For the best risk-adjusted performance (highest Sharpe, lowest drawdown)

Signal Logic:

- **Step 1 - Sentiment Gate:** Check pre-market Up/Down votes
 - Down Votes > 60% => LONG signal
 - Up Votes > 60% => SHORT signal
 - Otherwise => STAY FLAT (insufficient conviction)
- **Step 2 - Technical Confirmation (SMA Filter):**
 - LONG + (Close > SMA20) => EXECUTE LONG
 - SHORT + (Close < SMA20) => EXECUTE SHORT
 - Signal disagrees with SMA regime => STAY FLAT (filter blocks)

+31.6%	+10.4%	5.64	-3.4%	64.3%	70
Total Ret	CAGR	Sharpe	Max DD	Win Rate	Total Trades

Alternative – S3. Contrarian + SMA (Higher Return, More Trades)

For maximum total return with more frequent trading (229 trades compared to 70):

Signal Logic:

- **Step 1 - Sentiment Gate:** Any Up/Down imbalance (no conviction threshold)
 - Down > Up => LONG signal
 - Up > Down => SHORT signal
- **Step 2 - SMA Filter:**
 - LONG + (Close > SMA20) => EXECUTE LONG
 - SHORT + (Close < SMA20) => EXECUTE SHORT
 - Signal disagrees with SMA regime => STAY FLAT (filter blocks)

+109.4%	+27.5%	3.76	-10.5%	59.8%	229
Total Ret	CAGR	Sharpe	Max DD	Win Rate	Total Trades

Both Recommendations Comparison

Strategy	Trades	Win Rate	Total Ret	CAGR	Sharpe	Max DD
Buy and Hold	-	-	+6.5%	+2.1%	0.21	-35.9%
S3.Contrarian (base)	552	50.7%	+9.9%	+3.1%	0.31	-22.0%
S4. Extreme Contrarian	196	51.5%	+8.1%	+2.6%	0.61	-16.6%
S3.Contrarian + SMA	229	59.8%	+109.4%	+27.5%	3.76	-10.5%
S4.Extreme Contrarian + SMA	70	64.3%	+31.6%	+10.4%	5.64	-3.4%